WSQ COURSE SLIDES

Fundamentals of ISO 14001 Environmental Management System (EMS)

WSQ Course Code: TGS-2023037592

Tertiary Infotech Academy Pte Ltd | UEN: 201200696W

Version v8.0 | 20 August 2026

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COURSE ADMINISTRATION

Welcome and course orientation

Two days | 16 hours | beginner level

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Digital Attendance (Mandatory)

1. Complete AM, PM and Assessment digital attendance for this WSQ-funded course.
2. Use the SSG digital-attendance QR code shown by the trainer or administrator.
3. Scan with your mobile phone and confirm the submission before leaving the page.
4. At least 75% attendance is required for assessment and funding eligibility.

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer | Tertiary Infotech Academy

ROLE
Principal Trainer and course developer

EXPERTISE
Management systems, process improvement and applied technology

DELIVERS
WSQ professional and technical programmes

FOUNDER
Tertiary Infotech / Tertiary Courses

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Let's know each other

1. Name, organisation and role
2. One environmental interaction in your work
3. A recurring environmental risk or data gap
4. What useful course success would look like

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HOUSEKEEPING

Ground rules

1. **Be present**
   Return on time and complete digital attendance.

2. **Participate**
   Ask, challenge and contribute evidence.

3. **Respect**
   One conversation at a time; disagree professionally.

4. **Protect information**
   Use fictional or sanitised workplace data.

5. **Stay evidence-led**
   Label assumptions and missing data.

6. **Work safely**
   Activities are tabletop analysis, not live chemical handling.

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TWO-DAY JOURNEY

Course roadmap

Day 1 - Plan

- EMS foundations and 2026 status
- Context, leadership and aspects
- Planning, support and controls

Day 2 - Check

- Emergency readiness
- Performance and compliance
- Internal audit and review

Close - Act

- Corrective action
- Continual improvement
- 2026 transition plan

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SCHEDULE

Lesson plan: Day 1

Morning

1. 9:00-9:30 | Administration and outcomes
2. 9:30-10:30 | EMS, ISO family and PDCA
3. 10:30-10:40 | Tea break
4. 10:40-1:00 | Context, aspects and Activities 1-2

Afternoon

1. 1:00-1:45 | Lunch
2. 1:45-3:30 | Planning and compliance
3. 3:30-3:40 | Tea break
4. 3:40-6:00 | Objectives, controls and Activities 3-4

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SCHEDULE

Lesson plan: Day 2

Morning

1. 9:00-10:20 | Operation and emergency readiness
2. 10:20-10:30 | Tea break
3. 10:30-12:30 | Activities 5-6
4. 12:30-1:15 | Lunch

Afternoon

1. 1:15-3:00 | Evaluation, audit and Activity 7
2. 3:00-3:10 | Tea break
3. 3:10-4:00 | Improvement and Activity 8
4. 4:00-6:00 | WA then Case Study

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BY THE END OF THIS COURSE

Learning outcomes

1

LO1

Identify environmental hazards and assess risks within the workplace context using an ISO 14001 EMS.

2

LO2

Determine actions for environmental accidents and report unsafe conditions or acts to relevant authorities.

3

LO3

Evaluate environmental hazards and oversee ISO 14001 EMS-related training and assessment activities.

4

LO4

Enhance the intended outcomes of environmental management through ISO 14001 EMS continual improvement.

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Knowledge alignment

K1-K3

1. K1: Purpose, benefits, structure and PDCA logic of an ISO 14001 environmental management system
2. K2: Organisation context, interested parties, EMS scope, leadership, environmental policy and responsibilities
3. K3: Environmental hazards, aspects, impacts, significance, risks and opportunities, and compliance obligations

K4-K6

1. K4: Objectives, support, competence, communication, documented information, life-cycle perspective and operational control
2. K5: Monitoring, measurement, analysis, compliance evaluation, internal audit and management review
3. K6: Emergency response, nonconformity, corrective action, continual improvement and the ISO 14001:2026 changes

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Abilities practised through activities

A1-A4

1. A1: Identify environmental hazards and environmental aspects from workplace evidence
2. A2: Assess environmental impacts, significance, risks and opportunities using defined criteria
3. A3: Determine proportionate actions for environmental incidents and emergency situations
4. A4: Escalate and report environmental unsafe conditions, acts and compliance concerns

A5-A8

1. A5: Evaluate environmental hazards, operational controls and environmental performance evidence
2. A6: Plan and oversee EMS competence, awareness, training and evaluation activities
3. A7: Identify and prioritise opportunities to improve EMS performance and intended outcomes
4. A8: Document corrective action, effectiveness review and continual improvement according to organisational procedures

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Briefing for Assessment

1. Individual and open book: course slides, Learner Guide and approved activity materials only.
2. Place phones and other materials under the table; do not photograph or record scripts.
3. No discussion; answer in your own words using black or blue pen for hard-copy papers.
4. No correction fluid or tape; submit the completed paper on the LMS when instructed.
5. Re-assessment and appeal routes are available under the assessment policy.

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Assessment and funding

Written Assessment

- 6 open-ended SAQs
- K1-K6
- 1 hour | open book

Case Study

- 4 scenario questions
- A1-A8
- 1 hour | open book

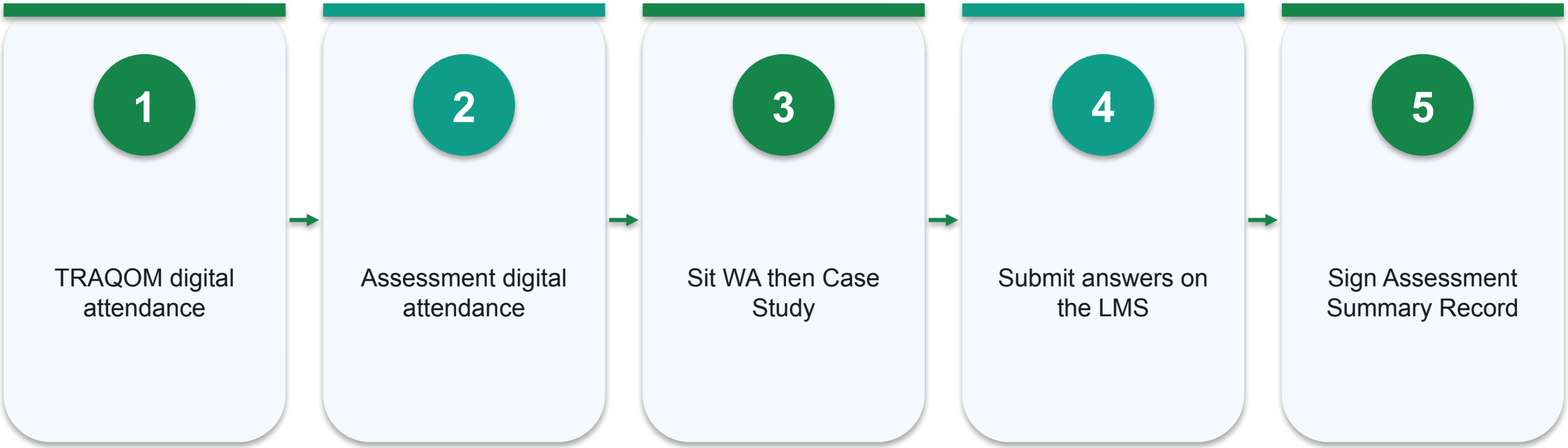
Competency

- Competent in both instruments
- At least 75% attendance
- Funding criteria apply

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ON ASSESSMENT DAY

Assessment flow



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Access the course activities

Option A - Git clone

1. Clone the repository to keep the folder structure.
2. Use the activity README/PDF and workbook together.
3. Keep your completed evidence in a separate working copy.

Option B - Download ZIP

1. Open the repository in a browser.
2. Choose Code, then Download ZIP.
3. Extract the ZIP before opening Excel workbooks.

Open public GitHub courseware →

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FOUNDATIONS

Environmental management that changes real outcomes

ISO 14001:2026 | EMS purpose | PDCA | evidence | certification

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ENVIRONMENTAL MANAGEMENT

An EMS is not a green binder.

It is how the organisation identifies interactions, controls risks, meets obligations and improves environmental performance.

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FAMILY OVERVIEW

The ISO 14000 family

EMS

- ISO 14001 requirements
- ISO 14004 guidance
- ISO 14005 phased implementation

Performance and claims

- ISO 14031 evaluation
- ISO 14020 series labels
- ISO 14064 series GHG

Life cycle and footprints

- ISO 14040/14044 LCA
- ISO 14046 water footprint
- ISO 14067 product carbon footprint

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CURRENT EDITION

Current standard status - 20 August 2026

1. ISO 14001:2026, Environmental management systems - Requirements with guidance for use
2. Edition 4 was published on 15 April 2026.
3. ISO 14001:2015 and ISO 14001:2015/Amd 1:2024 are withdrawn; their climate considerations are incorporated into the current edition.
4. Existing certified organisations should confirm transition arrangements with their accredited certification body and manage the gap review as controlled change.

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2026 EDITION

What ISO 14001:2026 sharpens

1. **Environmental conditions**
   More explicit attention to pollution, resources, climate, biodiversity and ecosystem health.

2. **Leadership**
   Stronger accountability and support for all relevant roles.

3. **Life-cycle scope**
   Life-cycle perspective is clearer when setting scope and managing the value chain.

4. **Planned change**
   New Clause 6.3 formalises planning and managing EMS-affecting changes.

5. **Emergencies**
   Consider all identified potential emergency situations.

6. **Governance**
   Clearer audit objectives, review results and measurable outcomes.

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Environmental management: three linked lenses

Inside-out impacts

- How activities affect air, water, land
- Resources, climate and biodiversity
- Normal, abnormal and emergency

Outside-in conditions

- Climate and resource constraints
- Ecosystem and pollution conditions
- How they affect EMS outcomes

Management-system response

- Governance and obligations
- Controls, competence and evidence
- Evaluation and improvement

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Aspect, impact and environmental hazard

Aspect and impact

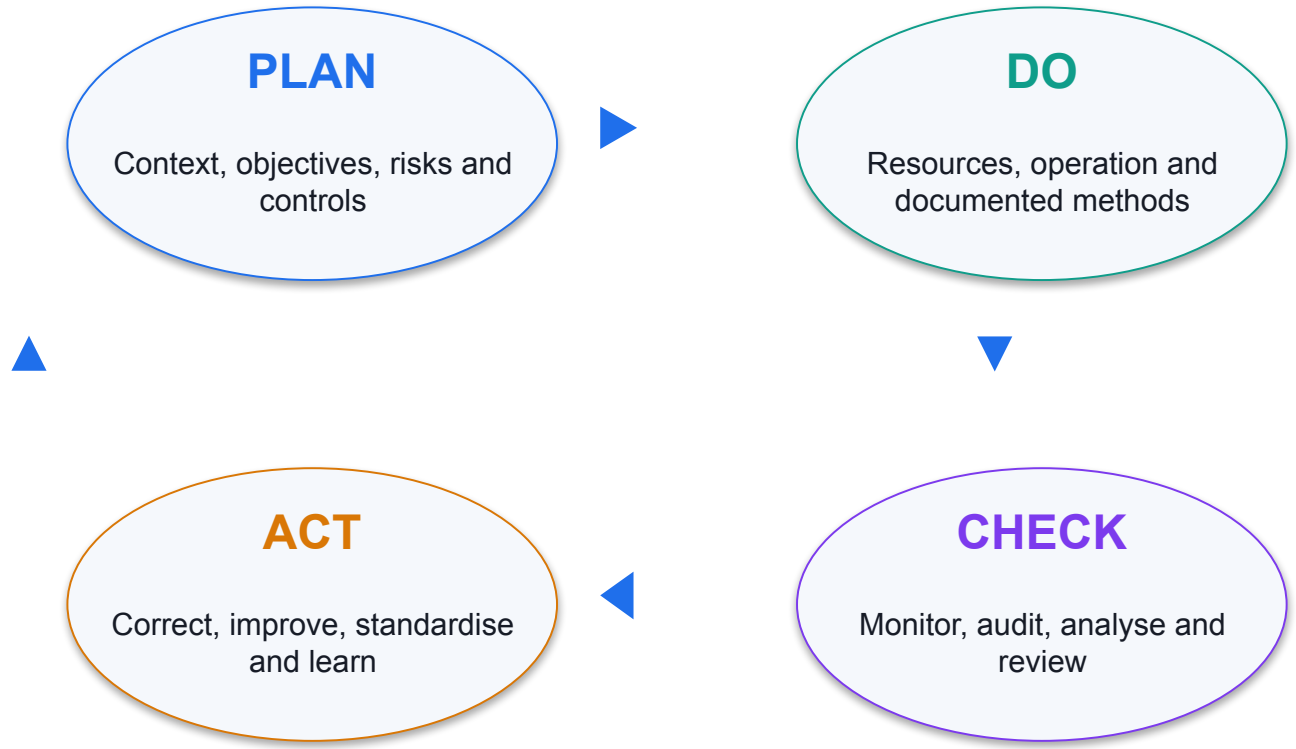
1. Aspect: an element that can interact with the environment
2. Impact: the resulting adverse or beneficial environmental change
3. Example: electricity use -> resource use and GHG emissions

Hazard and risk

1. Hazard/source: inherent potential to cause harm
2. Risk considers uncertainty, pathways, likelihood and consequence
3. Do not label every hazard as a significant aspect without criteria

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PDCA: the operating rhythm of an EMS



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ISO 14001:2026 structure: context to improvement



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EMS and sustainability reporting answer different questions

What the EMS does

1. EMS controls processes and environmental performance
2. ISO 14001 certification audits a management system
3. Evidence supports disciplined decisions and claims

What it does not prove

1. A sustainability report communicates selected disclosures
2. A carbon inventory answers a narrower measurement question
3. Certification does not certify every product or legal status

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BENEFITS

Why organisations use ISO 14001

Environmental value

- Prevent pollution and incidents
- Improve resource efficiency
- Address climate and biodiversity

Operational value

- Clarify controls and ownership
- Reduce waste and disruption
- Improve supplier oversight

Trust and access

- Support obligations and tenders
- Strengthen stakeholder confidence
- Provide auditable evidence

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Certification: what it is and is not

Voluntary framework

- ISO does not require certification
- Customers/tenders may require it
- EMS can operate without certification

Independent assessment

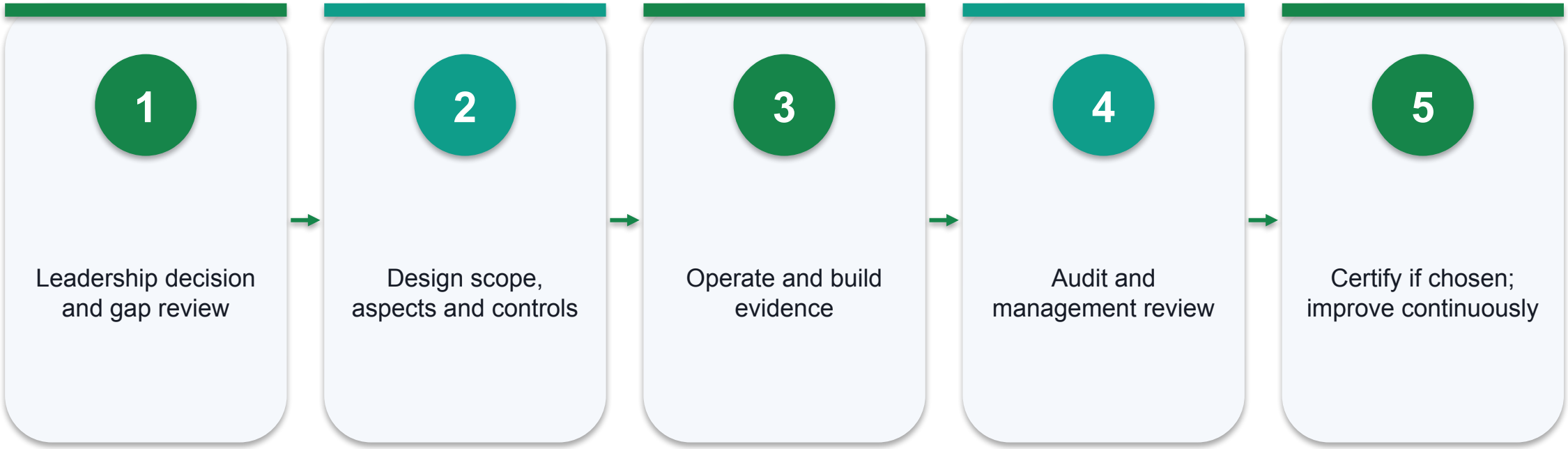
- ISO does not certify organisations
- Certification bodies audit
- Accreditation adds confidence

Ongoing cycle

- Stage 1 readiness
- Stage 2 implementation
- Surveillance and recertification

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Typical implementation and certification journey



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Integrated management systems

Shared architecture

1. Common harmonised structure
2. Shared context, leadership and planning
3. Common audit, review and improvement
4. Less duplication and clearer ownership

Examples

1. ISO 9001 | quality
2. ISO 14001 | environment
3. ISO 45001 | occupational health and safety
4. ISO/IEC 27001 | information security

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FICTIONAL CASE COMPANY

Meridian EcoPack: the connected case

1

Business

a fictional Singapore SME with 120 employees producing printed corrugated and fibre-based packaging for regional food, electronics and healthcare customers.

2

Customer signal

Two key customers require evidence of an ISO 14001:2026-aligned EMS and stronger value-chain environmental controls before contract renewal.

3

Evidence pattern

Electricity intensity rose 11%; mixed production waste increased; an ink tote spill nearly reached a drain; wastewater and contractor records are incomplete; nearby tenants reported intermittent odour; supplier environmental evidence varies.

4

Course mission

Build an evidence-led EMS that protects the environment, meets applicable obligations, improves performance and supports certification readiness.

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PROGRESSIVE EVIDENCE JOURNEY

Eight activities close one PDCA loop

1. **A1** — Build the Environmental Baseline and Aspect Inventory
2. **A2** — Assess Significance, Risks and Opportunities
3. **A3** — Build the Compliance Obligations Register
4. **A4** — Set Environmental Objectives and Build Competence
5. **A5** — Design Life-Cycle Operational Controls
6. **A6** — Run an Environmental Emergency Tabletop
7. **A7** — Evaluate Performance and Conduct an Internal Audit
8. **A8** — Close Corrective Action and Plan the 2026 Transition

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TOPIC 1

Overview of ISO 14001 EMS

ISO 14001:2026 - EMS purpose - PDCA - context - leadership - aspects and impacts

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CLAUSE 4.1

4.1 Organisation context and environmental conditions: intent and control

1

Intent

Determine issues and environmental conditions that affect the EMS and its intended outcomes.

2

Control 1

Consider pollution, resource availability, climate, biodiversity and ecosystem health.

3

Control 2

Connect internal and external issues to environmental and business decisions.

4

Control 3

Review context when operations, conditions or interested-party expectations change.

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CLAUSE 4.1

4.1 Organisation context and environmental conditions: evidence check

Useful objective evidence

1. Context and environmental-conditions register
2. Strategic and risk review evidence
3. Climate/biodiversity/resource scan
4. Management-review updates

Common weak practice

1. Static SWOT with no environmental meaning
2. Inside-out impacts considered but outside-in conditions missed
3. Climate relevance asserted without evidence
4. No action owner or review trigger

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CLAUSE 4.2

4.2 Interested parties and relevant requirements: intent and control

1

Intent

Identify relevant parties, needs and expectations, including those that become compliance obligations.

2

Control 1

Select parties because they affect or are affected by environmental performance.

3

Control 2

Separate expectations from mandatory or voluntarily accepted obligations.

4

Control 3

Monitor changes in regulator, community, customer, worker and value-chain needs.

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CLAUSE 4.2

4.2 Interested parties and relevant requirements: evidence check

Useful objective evidence

1. Interested-party matrix
2. Customer and permit requirements
3. Community/landlord communications
4. Review records and change alerts

Common weak practice

1. Listing every possible party
2. Treating every wish as binding
3. Ignoring affected communities or suppliers
4. No route from need to EMS control

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CLAUSE 4.3

4.3 Scope of the EMS: intent and control

1

Intent

Set a truthful EMS boundary that reflects context, activities, authority and life-cycle perspective.

2

Control 1

State sites, units, activities, products and services covered.

3

Control 2

Consider upstream and downstream stages that can be controlled or influenced.

4

Control 3

Make the scope available as controlled documented information.

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CLAUSE 4.3

4.3 Scope of the EMS: evidence check

Useful objective evidence

1. Approved scope statement
2. Site/process/value-chain boundary map
3. Authority and influence rationale
4. Alignment with certification scope

Common weak practice

1. Scope excludes a significant activity without rationale
2. Outsourcing treated as outside responsibility
3. Marketing scope broader than reality
4. Life-cycle perspective absent

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CLAUSE 4.4

4.4 Environmental management system: intent and control

1

Intent

Establish, implement, maintain and continually improve interacting EMS processes.

2

Control 1

Translate clauses into operational and governance processes.

3

Control 2

Define sequence, interaction, ownership, resources and evidence.

4

Control 3

Integrate environmental management into core business processes.

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4.4 Environmental management system: evidence check

Useful objective evidence

1. EMS process architecture
2. Roles and process ownership
3. Operational and performance records
4. Improvement and change evidence

Common weak practice

1. A document library mistaken for a system
2. EMS isolated within one department
3. Controls not connected to aspects
4. No evidence that processes achieve outcomes

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CLAUSE 5.1

5.1 Leadership and commitment: intent and control

1

Intent

Top management is accountable for EMS effectiveness and environmental performance.

2

Control 1

Align environmental direction with strategy and business processes.

3

Control 2

Provide resources and support all relevant roles in contributing to EMS results.

4

Control 3

Promote improvement and credible environmental decision-making.

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CLAUSE 5.1

5.1 Leadership and commitment: evidence check

Useful objective evidence

1. Leadership decisions and resources
2. Management participation in reviews
3. Environmental objectives linked to strategy
4. Evidence barriers were removed

Common weak practice

1. EMS delegated to the environmental coordinator
2. Policy promises without resources
3. Performance data not on leadership agenda
4. Support limited to management roles

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CLAUSE 5.2

5.2 Environmental policy: intent and control

1

Intent

Provide direction and commitments appropriate to purpose, context and environmental impacts.

2

Control 1

Commit to environmental protection, including pollution prevention and relevant context-specific commitments.

3

Control 2

Commit to meeting compliance obligations and continual improvement.

4

Control 3

Communicate, maintain and make the policy available.

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5.2 Environmental policy: evidence check

Useful objective evidence

1. Approved current policy
2. Objectives derived from commitments
3. Interview and communication evidence
4. Availability to relevant interested parties

Common weak practice

1. Generic green claims
2. Protection commitments unrelated to aspects
3. People can recite but not apply the policy
4. Policy not reflected in objectives

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CLAUSE 5.3

5.3 Roles, responsibilities and authorities: intent and control

1

Intent

Make EMS accountability, authority and reporting routes clear.

2

Control 1

Assign ownership for significant aspects, controls and compliance evaluation.

3

Control 2

Ensure EMS performance and improvement needs reach top management.

4

Control 3

Maintain cover during leave, contractor work and organisational change.

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CLAUSE 5.3

5.3 Roles, responsibilities and authorities: evidence check

Useful objective evidence

1. Role profiles and RACI
2. Escalation and incident routes
3. EMS performance reporting
4. Delegation and backup arrangements

Common weak practice

1. Everyone responsible means nobody owns it
2. Responsibility without authority
3. Contractor roles unclear
4. Environmental reporting stops at department level

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CLAUSE 6.1.1

6.1.1 Risks and opportunities - general: intent and control

1

Intent

Use context, parties, scope, aspects and obligations to protect intended EMS outcomes.

2

Control 1

Distinguish environmental impacts from risks/opportunities to the EMS and organisation.

3

Control 2

Address undesired effects and pursue beneficial opportunities.

4

Control 3

Integrate actions into EMS processes and evaluate effectiveness.

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CLAUSE 6.1.1

6.1.1 Risks and opportunities - general: evidence check

Useful objective evidence

1. Risk and opportunity register
2. Cause-action-owner linkage
3. Implemented controls and projects
4. Effectiveness review

Common weak practice

1. Risk register isolated from operations
2. Positive impact called an opportunity without business/system effect
3. Actions copied from templates
4. Completion confused with effectiveness

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CLAUSE 6.1.2

6.1.2 Environmental aspects: intent and control

1

Intent

Identify aspects and impacts across a life-cycle perspective and determine significance using established criteria.

2

Control 1

Consider normal, abnormal and all potential emergency situations.

3

Control 2

Cover activities, products, services, planned changes and new developments.

4

Control 3

Communicate significant aspects to relevant levels and functions.

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CLAUSE 6.1.2

6.1.2 Environmental aspects: evidence check

Useful objective evidence

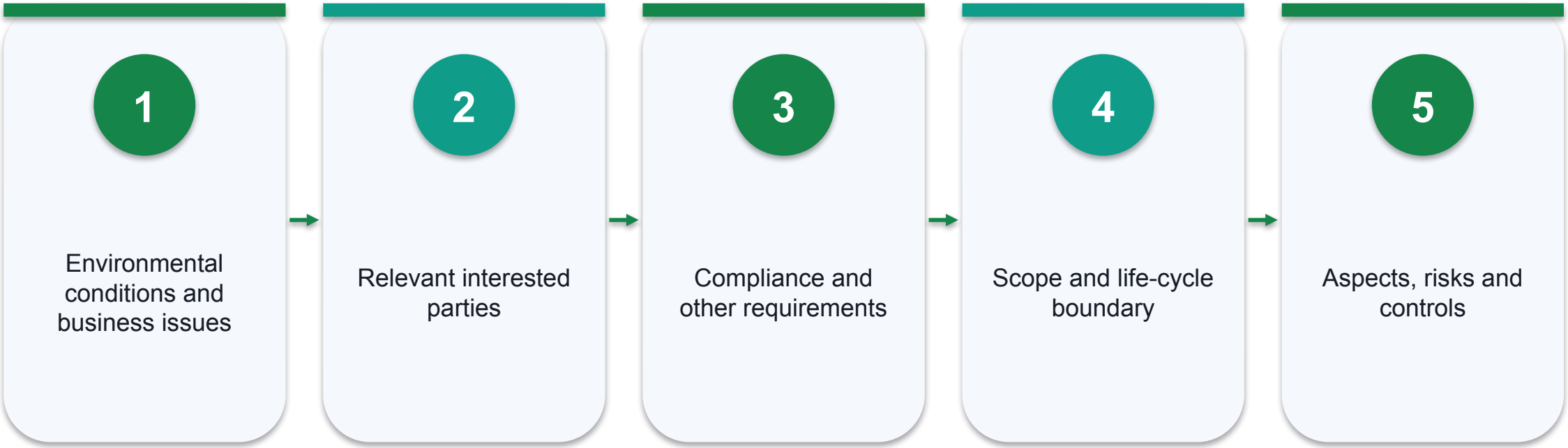
1. Aspect-impact register
2. Significance method and rationale
3. Life-cycle map
4. Change and emergency-scenario review

Common weak practice

1. Aspect and impact reversed
2. Only normal operations considered
3. Scores without evidence or criteria
4. Significant aspects not translated into controls

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Context creates the line of sight



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Environmental conditions to test

1. **Pollution**
   Existing air, water, soil, noise or contamination conditions.

2. **Resources**
   Availability and quality of water, energy, materials and land.

3. **Climate**
   Physical and transition conditions affecting operations/value chain.

4. **Biodiversity**
   Sensitive habitats, species and ecosystem dependencies.

5. **Community**
   Neighbour exposure, amenity and environmental justice concerns.

6. **Change**
   New products, capacity, technology, suppliers or sites.

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Direct control and organisational influence

Direct control

1. On-site process settings
2. Chemical storage and drain protection
3. Maintenance and monitoring
4. Emergency response

Influence

1. Supplier specifications
2. Contractor requirements
3. Customer use information
4. End-of-life design and recovery

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Build the Environmental Baseline and Aspect Inventory

ACTIVITY 1

Trace MEP's materials, energy, water, emissions, waste and emergency interfaces and distinguish hazard, aspect and impact.

EXPECTED OUTPUT

A process-boundary map, environmental baseline and aspect-impact inventory with evidence references.

Resources: Scenario guide PDF, EMS Evidence Workbook (Excel), site-plan evidence cards

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ACTIVITY 1 | DECISION PROMPT

Build the Environmental Baseline and Aspect Inventory



MEP is preparing its first ISO 14001:2026-aligned EMS. Managers have energy bills and waste invoices, but no shared view of where activities interact with air, water, land, resources or neighbours.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 1 | VERIFY

Build the Environmental Baseline and Aspect Inventory

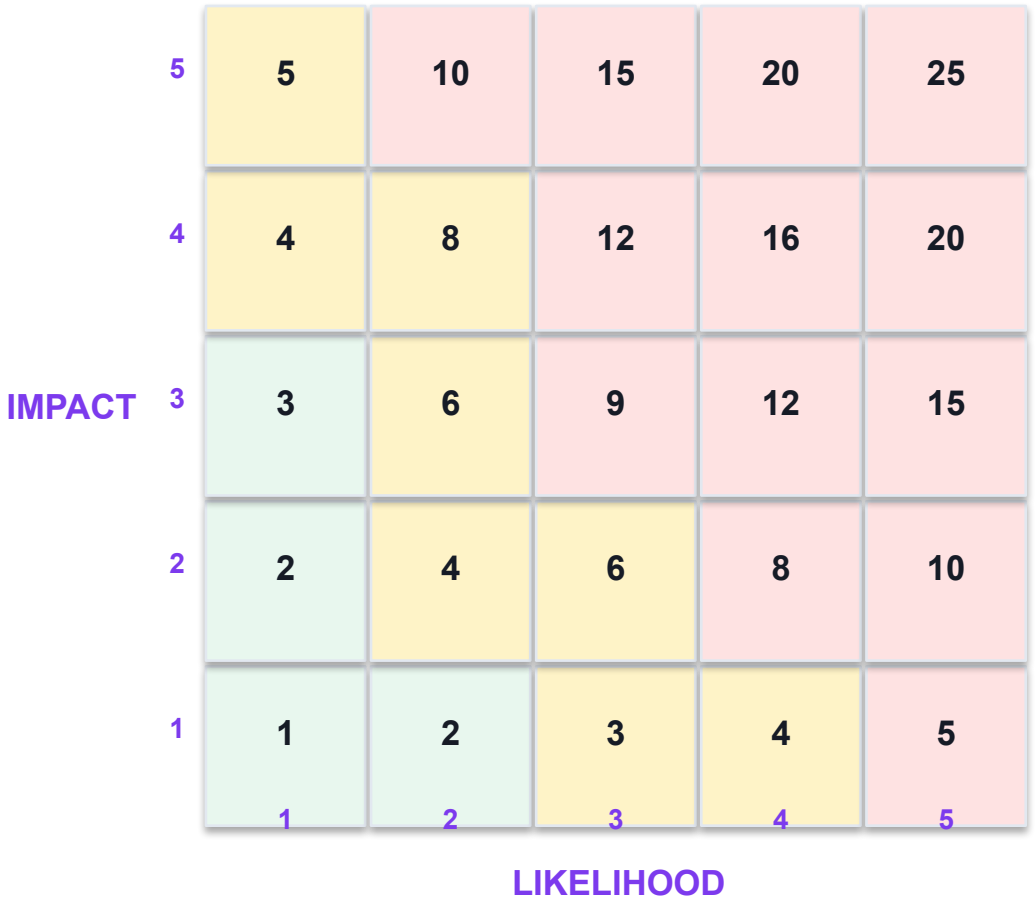
CHECK YOUR EVIDENCE

The inventory covers normal, abnormal and potential emergency conditions; every aspect-impact pair is logically linked and traceable to supplied evidence or an explicit evidence gap.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Risk-based thinking: prioritise proportionately



Green = lower | Amber = moderate | Red = higher

Determine What could affect the intended result?

Plan Actions proportionate to potential impact

Integrate Embed actions in real processes

Evaluate Check whether actions are effective

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CLAUSE 6.1

Significance and EMS risk answer different questions

Aspect significance

1. How material is the environmental interaction/impact?
2. Use defined significance criteria
3. Consider conditions and mandatory overrides
4. Communicate significant aspects

Risks and opportunities

1. What uncertainty affects EMS outcomes or the organisation?
2. Connect context, parties, aspects and obligations
3. Plan proportionate action
4. Evaluate action effectiveness

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Assess Significance, Risks and Opportunities

ACTIVITY 2

Score MEP's aspects using severity, likelihood, compliance exposure, stakeholder concern and control effectiveness, then test sensitivity.

EXPECTED OUTPUT

A significance matrix, prioritised aspects register and risk/opportunity action shortlist.

Resources: Significance and Risk Workbook (Excel), scoring calibration checklist PDF

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ACTIVITY 2 | DECISION PROMPT

Assess Significance, Risks and Opportunities



MEP's departments use different rating scales. Production prioritises cost, Facilities prioritises spill prevention, and Sales prioritises customer carbon requests. Leadership needs one defensible method.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 2 | VERIFY

Assess Significance, Risks and Opportunities

CHECK YOUR EVIDENCE

Priority aspects are produced by the approved criteria and override rules; scores have evidence/rationale, uncertainties are visible and opportunity actions are not confused with positive impacts.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Lunch Break

45 minutes | return ready for Topic 2

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2

TOPIC 2

Planning, support and operation

Risks and opportunities - objectives - competence - information - controls - emergency readiness

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CLAUSE 6.1.3

6.1.3 Compliance obligations: intent and control

1

Intent

Determine, access, understand and account for applicable legal and other obligations.

2

Control 1

Link each obligation to relevant aspects, processes and controls.

3

Control 2

Keep authoritative sources, applicability rationale and change monitoring current.

4

Control 3

Maintain knowledge of status without unsupported legal conclusions.

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CLAUSE 6.1.3

6.1.3 Compliance obligations: evidence check

Useful objective evidence

1. Obligations register with current sources
2. Permits, licences and contract commitments
3. Owners, evidence and review dates
4. Change-monitoring and escalation records

Common weak practice

1. A list of law titles only
2. Outdated copied requirements
3. No applicability rationale
4. Certification mistaken for legal compliance

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CLAUSE 6.1.4

6.1.4 Planning action: intent and control

1

Intent

Plan how significant aspects, obligations, risks and opportunities will be addressed and evaluated.

2

Control 1

Select proportionate action and integrate it into operational or business processes.

3

Control 2

Consider technological options and financial/operational requirements.

4

Control 3

Define how effectiveness will be evaluated.

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CLAUSE 6.1.4

6.1.4 Planning action: evidence check

Useful objective evidence

1. Integrated action plans
2. Control and project changes
3. Owners, resources and milestones
4. Effectiveness criteria and results

Common weak practice

1. Separate registers with no action
2. Every item given the same response
3. Technology chosen before need
4. Actions closed on date alone

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CLAUSE 6.2

6.2 Environmental objectives and plans: intent and control

1

Intent

Translate policy and priorities into measurable, monitored and communicated results.

2

Control 1

Set objectives considering significant aspects, obligations, risks and opportunities.

3

Control 2

Define what, resources, owner, completion date and evaluation method.

4

Control 3

Use baselines, boundaries, data-quality rules and response thresholds.

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6.2 Environmental objectives and plans: evidence check

Useful objective evidence

1. Objective and KPI dictionary
2. Baseline and target rationale
3. Action plans and resources
4. Trend, decisions and updates

Common weak practice

1. Reduce waste without baseline
2. Absolute target hides production change
3. KPI without a management response
4. Activity completed but result unmeasured

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CLAUSE 6.3

6.3 Planning and managing changes: intent and control

1

Intent

Plan and manage changes that affect or can affect intended EMS outcomes.

2

Control 1

Identify the purpose, consequences and environmental/compliance risks of change.

3

Control 2

Coordinate resources, responsibilities, controls, competence and communication.

4

Control 3

Verify implementation and results, including unintended effects.

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6.3 Planning and managing changes: evidence check

Useful objective evidence

1. EMS change request and impact review
2. Control/competence/document updates
3. Implementation and rollback evidence
4. Post-change effectiveness review

Common weak practice

1. Project change bypasses EMS review
2. Temporary control becomes permanent
3. Supplier or process interfaces missed
4. Change closed at installation rather than result

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CLAUSE 7.1

7.1 Resources: intent and control

1

Intent

Provide people, infrastructure, technology, finance and external support needed for the EMS.

2

Control 1

Base resource decisions on risk, objectives and performance evidence.

3

Control 2

Maintain monitoring equipment, containment and emergency capability.

4

Control 3

Address resource gaps through management decisions.

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CLAUSE 7.1

7.1 Resources: evidence check

Useful objective evidence

1. Budget and resource plan
2. Maintenance/calibration records
3. Emergency equipment inspections
4. Management decisions on gaps

Common weak practice

1. Resources approved without needs analysis
2. Critical equipment unavailable
3. Monitoring validity unprotected
4. EMS depends on one person

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CLAUSE 7.2

7.2 Competence: intent and control

1

Intent

Ensure people doing work under organisational control are competent where environmental performance is affected.

2

Control 1

Define competence by role, task, aspect and risk.

3

Control 2

Take action to acquire competence and evaluate the action's effectiveness.

4

Control 3

Retain appropriate evidence, including contractor competence.

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CLAUSE 7.2

7.2 Competence: evidence check

Useful objective evidence

1. Role-competence matrix
2. Observed task/drill results
3. Qualifications and supervised practice
4. Gap and refresher actions

Common weak practice

1. Attendance treated as competence
2. Generic annual training
3. No evaluation after change
4. Contractors excluded

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CLAUSE 7.3

7.3 Awareness: intent and control

1

Intent

Ensure people understand policy, significant aspects, contribution, benefits and consequences of nonconformity.

2

Control 1

Make awareness relevant to actual tasks and decisions.

3

Control 2

Explain how individual work changes impacts and EMS results.

4

Control 3

Confirm understanding through dialogue or observation.

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CLAUSE 7.3

7.3 Awareness: evidence check

Useful objective evidence

1. Task briefings and interviews
2. Toolbox talks tied to current risks
3. Observed behaviours
4. Incident and improvement participation

Common weak practice

1. Poster-only awareness
2. People know slogans but not controls
3. Consequences framed only as punishment
4. No update after aspect/control change

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CLAUSE 7.4

7.4 Communication: intent and control

1

Intent

Plan consistent internal and external communication and ensure information is reliable.

2

Control 1

Define what, when, with whom, how and who communicates.

3

Control 2

Account for compliance obligations and respond to relevant communications.

4

Control 3

Keep communicated information consistent with EMS evidence.

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7.4 Communication: evidence check

Useful objective evidence

1. Communication plan and log
2. Authority/community/customer records
3. Incident notification matrix
4. Approved performance disclosures

Common weak practice

1. Broadcast email treated as communication
2. Unverified green claims
3. External queries lack ownership
4. Notification routes obsolete

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CLAUSE 7.5

7.5 Documented information: intent and control

1

Intent

Create and control information needed to operate the EMS and demonstrate results.

2

Control 1

Identify, format, review and approve information appropriately.

3

Control 2

Make current information available and protect integrity/confidentiality.

4

Control 3

Control distribution, access, changes, retention and external-origin documents.

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CLAUSE 7.5

7.5 Documented information: evidence check

Useful objective evidence

1. Controlled registers and procedures
2. Point-of-use current information
3. Access/change history
4. Retention and disposal evidence

Common weak practice

1. Documentation for its own sake
2. Obsolete emergency contacts
3. Records editable without traceability
4. External permits/SDS uncontrolled

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CLAUSE 8.1

8.1 Operational planning and control: intent and control

1

Intent

Control significant aspects and obligations across operations and the relevant value chain.

2

Control 1

Establish operating criteria and control processes, including planned changes.

3

Control 2

Control or influence externally provided processes, products and services proportionately.

4

Control 3

Use life-cycle perspective in design, procurement, delivery, use and end-of-life information.

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CLAUSE 8.1

8.1 Operational planning and control: evidence check

Useful objective evidence

1. Operational control plans
2. Work/maintenance/contractor records
3. Supplier requirements and evaluations
4. Change and effectiveness evidence

Common weak practice

1. Procedure lacks operating criteria
2. Outsourcing treated as transferred accountability
3. Control relies only on training
4. Downstream influence ignored

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CLAUSE 8.2

8.2 Emergency preparedness and response: intent and control

1

Intent

Prepare for and respond to all identified potential emergency situations.

2

Control 1

Prevent or mitigate environmental consequences and respond to actual events.

3

Control 2

Test arrangements periodically where practicable and evaluate performance.

4

Control 3

Review after emergencies/tests and communicate relevant information.

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CLAUSE 8.2

8.2 Emergency preparedness and response: evidence check

Useful objective evidence

1. Scenario and response plans
2. Current contacts and site/drain plans
3. Drill observations and timed results
4. Post-event corrective actions

Common weak practice

1. Only fire scenarios considered
2. Plans not matched to pathways/receptors
3. Drill attendance without objectives
4. Contacts and equipment not verified

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CLAUSE 6.1.3

Compliance obligation workflow

Build the register

1. Find current authoritative sources
2. Decide applicability with competent input
3. Translate obligation into process/control
4. Maintain owner and evidence

Maintain knowledge

1. Monitor changes
2. Evaluate status periodically
3. Correct noncompliance or evidence gaps
4. Escalate uncertainty rather than guess

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Build the Compliance Obligations Register

ACTIVITY 3

Screen MEP's activities against Singapore environmental obligation categories, customer commitments and permit conditions.

EXPECTED OUTPUT

A compliance obligations register, applicability rationale, evidence map and escalation log.

Resources: Compliance Register Workbook (Excel), obligation-screening checklist PDF, NEA source links

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Build the Compliance Obligations Register



MEP holds licences and service contracts in separate folders. A customer code, wastewater conditions, hazardous-substance storage controls and waste records are not linked to process owners or evidence.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 3 | VERIFY

Build the Compliance Obligations Register

CHECK YOUR EVIDENCE

Every register row has a source, applicability rationale, owner, evidence, review frequency and escalation route; unresolved interpretation is escalated rather than guessed.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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CLAUSE 6.2

Environmental objective design test

SMART and owned

1. Significant aspect or strategic outcome
2. Measurable definition and baseline
3. Target, boundary and due date
4. Owner, resources and action plan

Decision-ready

1. Relevant to policy and obligations
2. Leading and lagging indicators
3. Data-quality and review cadence
4. Response threshold and result evaluation

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Set Environmental Objectives and Build Competence

ACTIVITY 4

Create an electricity/waste objective set and a competence plan for operators, supervisors, contractors and emergency roles.

EXPECTED OUTPUT

A SMART objective/KPI plan, action register, role-competence matrix and evaluation design.

Resources: Objectives and Competence Workbook (Excel), objective-design and training-evaluation checklist PDF

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ACTIVITY 4 | DECISION PROMPT

Set Environmental Objectives and Build Competence



MEP's policy promises resource efficiency and pollution prevention, but targets have no baseline, owners or response thresholds. Annual awareness slides are delivered without checking competence.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 4 | VERIFY

Set Environmental Objectives and Build Competence

CHECK YOUR EVIDENCE

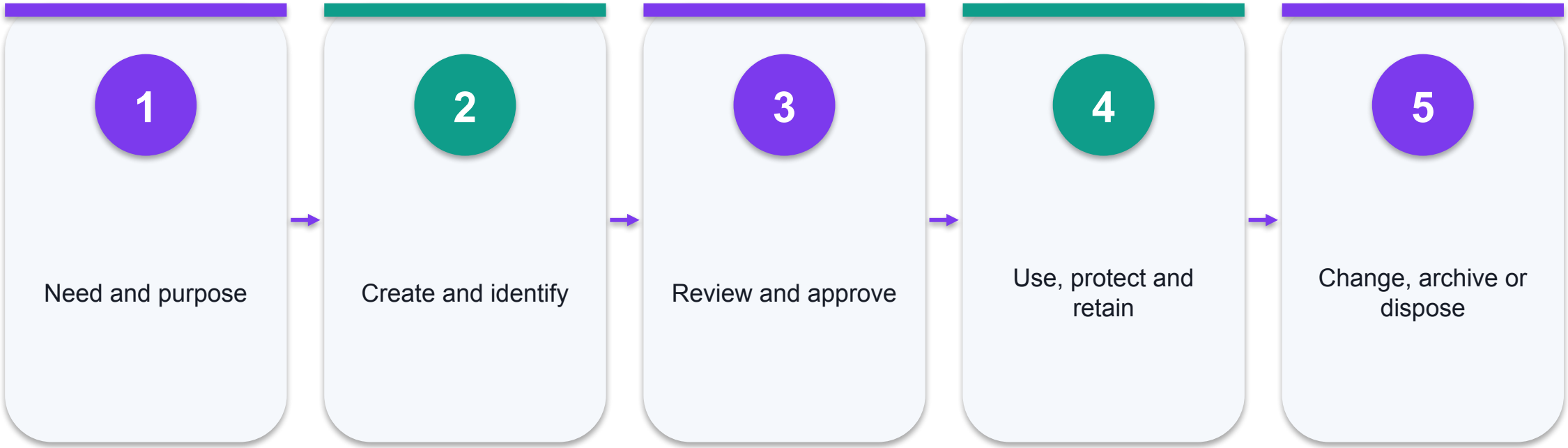
Each objective links to policy and a priority aspect; KPIs are reproducible and decision-ready; every critical role has competence criteria, evidence method, owner and review date.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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CLAUSE 7.5

Documented information life cycle



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CLAUSE 8.1

Operational-control hierarchy

Prevent or eliminate

- Avoid hazardous material/process
- Design out waste or releases
- Reduce demand at source

Engineer and contain

- Closed transfer and bunding
- Interlocks and treatment
- Physical segregation

Administer and respond

- Operating criteria and checks
- Competence and communication
- Emergency arrangements

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Life-cycle perspective in practice

Upstream and operations

1. Material and supplier selection
2. Design and production controls
3. Packaging and distribution
4. Contractor and outsourced work

Downstream

1. Use-phase information
2. Reuse, recovery and recycling
3. End-of-life impacts
4. Evidence where influence is limited

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Design Life-Cycle Operational Controls

ACTIVITY 5

Control ink, paper, cleaning, maintenance, waste handling and supplier interfaces using the hierarchy of control and life-cycle perspective.

EXPECTED OUTPUT

An operational control plan, supplier requirement matrix and control-verification schedule.

Resources: Operational Control Workbook (Excel), life-cycle and supplier checklist PDF

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Design Life-Cycle Operational Controls



MEP controls the printing line but has weak requirements for paper suppliers, cleaning contractors, waste collectors and transport providers. Instructions describe tasks but not environmental acceptance criteria.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 5 | VERIFY

Design Life-Cycle Operational Controls

CHECK YOUR EVIDENCE

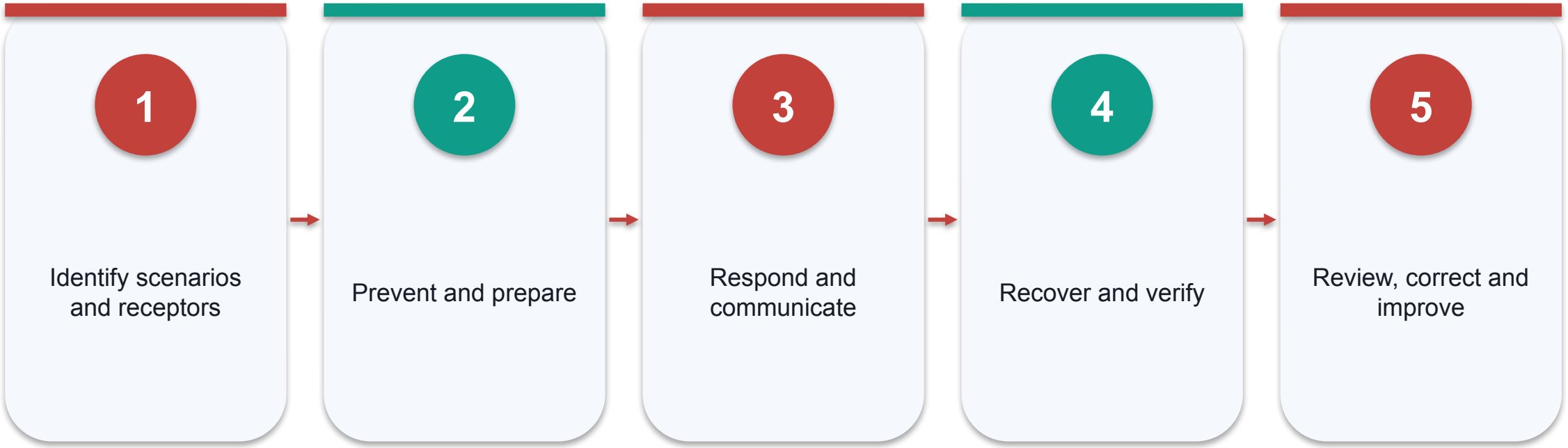
Controls address the actual cause of impact, include operating criteria and evidence, distinguish control from influence, and extend proportionately across relevant value-chain interfaces.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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CLAUSE 8.2

Emergency readiness: before, during and after



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Environmental and safety priorities interact

Human safety

1. Stop work and protect people
2. Use competent incident command
3. Control source only when safe
4. Coordinate emergency services

Environmental protection

1. Protect drains, land, air and neighbours
2. Preserve time-stamped evidence
3. Use approved notification decisions
4. Manage clean-up waste and recovery

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Run an Environmental Emergency Tabletop

ACTIVITY 6

Respond to an ink-tote spill near a storm-water drain while protecting people, the environment and evidence integrity.

EXPECTED OUTPUT

A time-stamped response log, notification decision record, containment plan and post-drill improvement actions.

Resources: Emergency Tabletop Workbook (Excel), spill-response role cards and observer checklist PDF

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ACTIVITY 6 | DECISION PROMPT

Run an Environmental Emergency Tabletop



At 10:20 a forklift punctures a 600-litre water-based ink tote. Product flows toward an internal drain. One operator raises the alarm; the quantity released, drain destination and regulatory notification threshold are initially uncertain.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Run an Environmental Emergency Tabletop

CHECK YOUR EVIDENCE

The team protects people and receptors, records decisions and uncertainties, uses the approved escalation path, preserves evidence and converts drill findings into owned improvement actions.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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CHECK YOUR UNDERSTANDING

Day 1 recap

1. **Context**
   Environmental conditions, interested parties and scope.

2. **Leadership**
   Policy, accountability and support for roles.

3. **Planning**
   Aspects, obligations, risks, objectives and change.

4. **Support**
   Resources, competence, awareness, communication and information.

5. **Operation**
   Life-cycle controls and value-chain oversight.

6. **Emergency**
   Credible scenarios, response, communication and learning.

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DAY 2

Performance evaluation and continual improvement

Emergency readiness, performance evaluation, audit, management review and continual improvement

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TOPIC 3

Performance evaluation

Monitoring - measurement - compliance evaluation - internal audit - management review

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CLAUSE 9.1.1

9.1.1 Monitoring, measurement, analysis and evaluation: intent and control

1

Intent

Use valid data to evaluate environmental performance and EMS effectiveness.

2

Control 1

Define what, methods, criteria, timing and evaluation responsibilities.

3

Control 2

Maintain calibrated/verified equipment where needed and assess data quality.

4

Control 3

Communicate relevant performance and retain evidence.

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CLAUSE 9.1.1

9.1.1 Monitoring, measurement, analysis and evaluation: evidence check

Useful objective evidence

1. KPI definitions and raw data
2. Meter/calibration and chain-of-custody records
3. Trend analysis with decisions
4. Performance communications

Common weak practice

1. Absolute data hides production change
2. Missing data silently interpolated
3. Dashboard without decisions
4. Activity count mistaken for impact reduction

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CLAUSE 9.1.2

9.1.2 Evaluation of compliance: intent and control

1

Intent

Periodically evaluate whether compliance obligations are met and maintain knowledge of status.

2

Control 1

Set frequency based on obligation, risk and change.

3

Control 2

Use competent evaluators and sufficient current evidence.

4

Control 3

Act on noncompliance or insufficient evidence and retain results.

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CLAUSE 9.1.2

9.1.2 Evaluation of compliance: evidence check

Useful objective evidence

1. Evaluation schedule and completed records
2. Permit/monitoring evidence
3. Status rationale and evaluator
4. Actions and re-evaluation

Common weak practice

1. Legal register mistaken for evaluation
2. Audit sample used as the only compliance check
3. Unknown marked compliant
4. No follow-up after a failed condition

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CLAUSE 9.2

9.2 Internal audit: intent and control

1

Intent

Provide objective information on conformity and effective implementation.

2

Control 1

Plan a risk-informed programme with documented objectives.

3

Control 2

Define scope, criteria, methods, responsibilities and impartiality.

4

Control 3

Report results, correct issues and follow up without undue delay.

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CLAUSE 9.2

9.2 Internal audit: evidence check

Useful objective evidence

1. Audit programme and objectives
2. Process-based plan/checklist
3. Traceable evidence and findings
4. Correction/corrective-action follow-up

Common weak practice

1. Every department audited equally regardless of risk
2. Auditor checks own work
3. Checklist tick treated as evidence
4. Finding prescribes a consultant solution

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CLAUSE 9.3

9.3 Management review: intent and control

1

Intent

Top management evaluates continuing suitability, adequacy, effectiveness and strategic alignment.

2

Control 1

Review context, aspects, obligations, objectives, performance, audits and resources as one story.

3

Control 2

Consider value-chain performance, changing conditions and improvement opportunities.

4

Control 3

Make and retain decisions on opportunities, changes and resources.

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CLAUSE 9.3

9.3 Management review: evidence check

Useful objective evidence

1. Decision-ready input pack
2. Minutes with results and rationale
3. Resource/change approvals
4. Owners and closure tracking

Common weak practice

1. Data presentation without decisions
2. Only good news shown
3. Environmental outcomes disconnected from business
4. Actions lack owners or resources

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CLAUSE 9.1

A useful environmental indicator has six parts

1. **Purpose**
   Which decision will it support?
2. **Boundary**
   Which site, process, product and period?
3. **Method**
   How is it collected, normalised and calculated?
4. **Quality**
   How are completeness, calibration and anomalies controlled?
5. **Owner**
   Who validates and responds?
6. **Threshold**
   When is intervention required?

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MEASUREMENT

Leading and lagging indicators

Leading signals

1. Control inspection completion
2. Competence demonstration
3. Preventive-maintenance adherence
4. Supplier-risk action closure

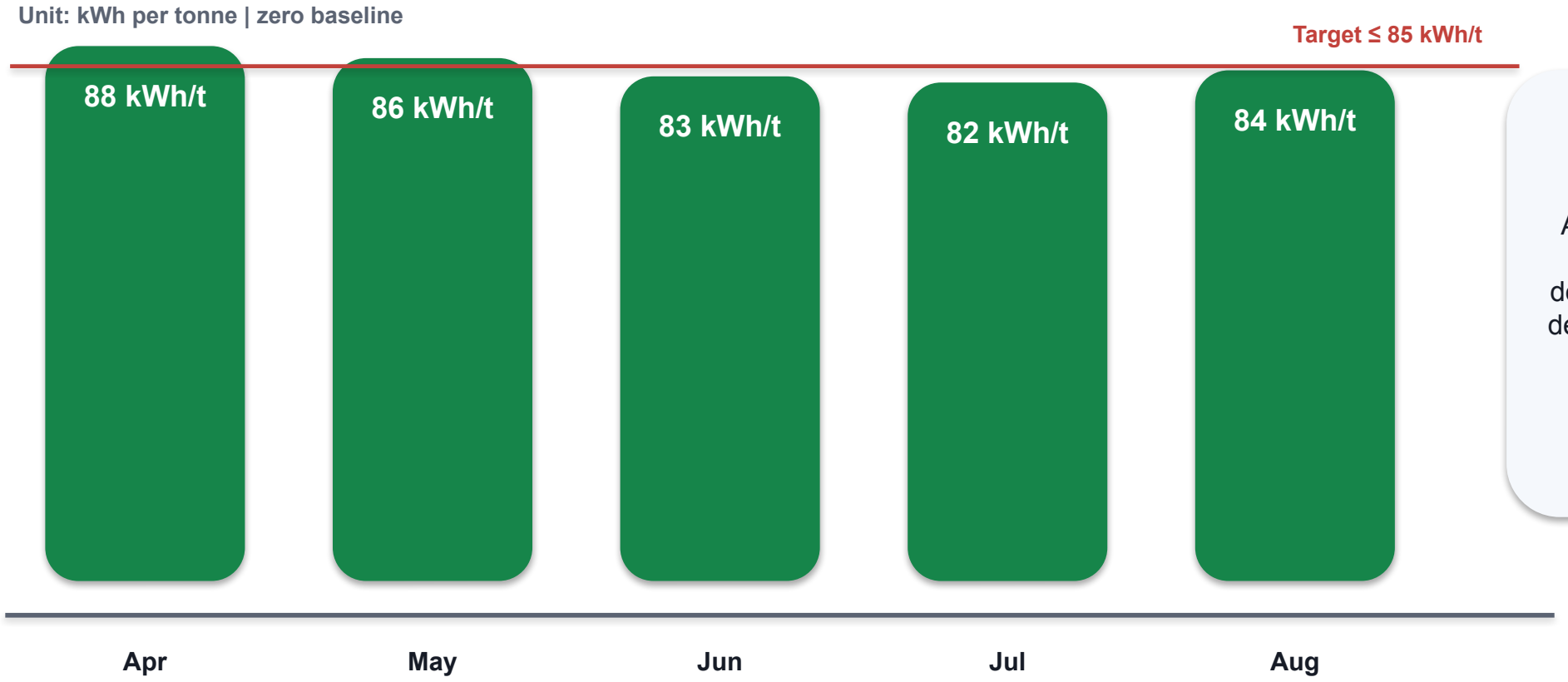
Lagging results

1. Energy and water intensity
2. Waste generation/diversion
3. Discharge/emission results
4. Incidents and complaints

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WHAT THE DATA SHOWS

MEP energy intensity - indicative trend



READ IT

A chart highlights patterns. A decision still needs definitions, context, valid data and a response threshold.

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THREE DIFFERENT CHECKS

Monitoring, compliance evaluation and audit

Performance and obligations

1. Monitoring: collect operational/performance evidence
2. Compliance evaluation: decide status against obligations
3. Both use defined competence, method and records

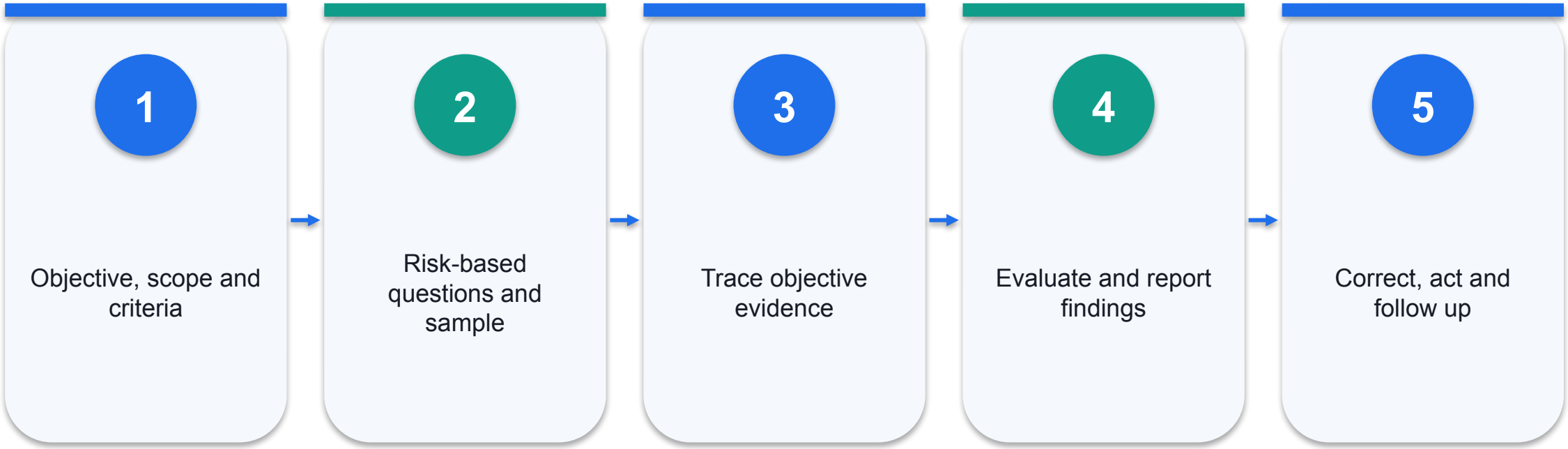
System assurance

1. Internal audit: evaluate EMS conformity and effectiveness
2. Certification audit: independent third-party assessment
3. None should claim legal compliance without sufficient evidence

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CLAUSE 9.2

Internal audit trail



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Audit evidence triangulation

Documents

- What should happen
- Current approved criteria
- Required records

Interviews

- What people understand
- How decisions are made
- Where variation occurs

Observation and records

- What actually happened
- Traceable samples
- Results and effectiveness

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Write a defensible nonconformity

Required structure

1. Criterion: the applicable requirement
2. Evidence: specific, traceable facts
3. Gap: how evidence fails the criterion
4. Scope: affected sample/process

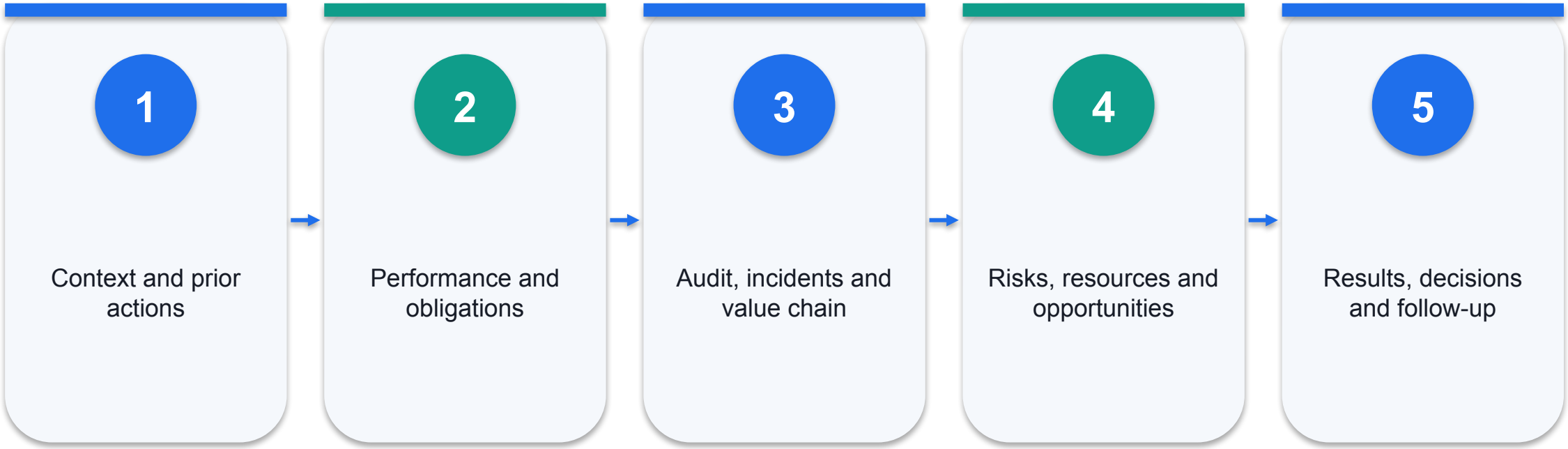
Avoid

1. Vague: environmental controls are poor
2. Opinion: staff seemed careless
3. Prescription: buy this software
4. Overreach: one sample proves total failure

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CLAUSE 9.3

Management review converts evidence into decisions



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Evaluate Performance and Conduct an Internal Audit

ACTIVITY 7

Analyse MEP's six-month KPI data, sample control records and conduct a process-based audit before management review.

EXPECTED OUTPUT

A KPI interpretation, compliance-evaluation record, audit plan, evidence log and defensible findings.

Resources: Performance and Audit Workbook (Excel), audit role-play evidence pack and checklist PDF

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Evaluate Performance and Conduct an Internal Audit



MEP reports that waste diversion improved, but production volume changed and two months of wastewater data are missing. The prior audit programme covered every department equally and did not state objectives.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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Evaluate Performance and Conduct an Internal Audit

CHECK YOUR EVIDENCE

Conclusions respect data limitations; compliance status is evidence-based; the audit trail is reproducible; every finding contains criterion, evidence and gap; training follow-up is tied to observed competence needs.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

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Lunch Break

45 minutes | return ready for Topic 4

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TOPIC 4

Continual improvement

Incident learning - nonconformity - root cause - corrective action - effectiveness - transition

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CLAUSE 10.1

10.1 Improvement - general: intent and control

1

Intent

Determine opportunities and implement actions needed to achieve intended EMS outcomes.

2

Control 1

Use performance, audit, incident and review evidence.

3

Control 2

Improve environmental performance as well as EMS processes.

4

Control 3

Prioritise by risk, obligation, impact and strategic value.

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10.1 Improvement - general: evidence check

Useful objective evidence

1. Improvement pipeline
2. Business/environmental case
3. Implementation evidence
4. Benefits and learning review

Common weak practice

1. Too many unprioritised ideas
2. Improvement means documentation only
3. No baseline or outcome
4. Local benefit transfers impact elsewhere

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10.2 Nonconformity and corrective action: intent and control

1

Intent

React, control consequences, eliminate causes and verify effectiveness.

2

Control 1

Correct and mitigate adverse environmental impacts.

3

Control 2

Determine cause and whether similar nonconformities exist or could occur.

4

Control 3

Implement action, review effectiveness and update EMS risks, controls or information.

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10.2 Nonconformity and corrective action: evidence check

Useful objective evidence

1. NCR and containment evidence
2. Root-cause verification
3. Cause-linked action plan
4. Effectiveness and system-update evidence

Common weak practice

1. Correction confused with corrective action
2. Root cause is person error
3. Similar situations not reviewed
4. Closure based on due date only

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CLAUSE 10.3

10.3 Continual improvement: intent and control

1

Intent

Continually improve EMS suitability, adequacy and effectiveness to enhance environmental performance.

2

Control 1

Use analysis and management-review results to select priorities.

3

Control 2

Standardise successful changes and preserve competence/evidence.

4

Control 3

Review residual impacts, unintended effects and next opportunities.

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10.3 Continual improvement: evidence check

Useful objective evidence

1. Improvement portfolio
2. Before/after environmental results
3. Updated standard work and competence
4. Lessons and next-review decision

Common weak practice

1. One-off campaign with no sustainment
2. Activity reported without outcome
3. Success claimed from incomplete data
4. Learning not applied across the system

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Three different responses

Correction

- Fix the detected problem
- Restore control or clean up
- Immediate condition focus

Containment/mitigation

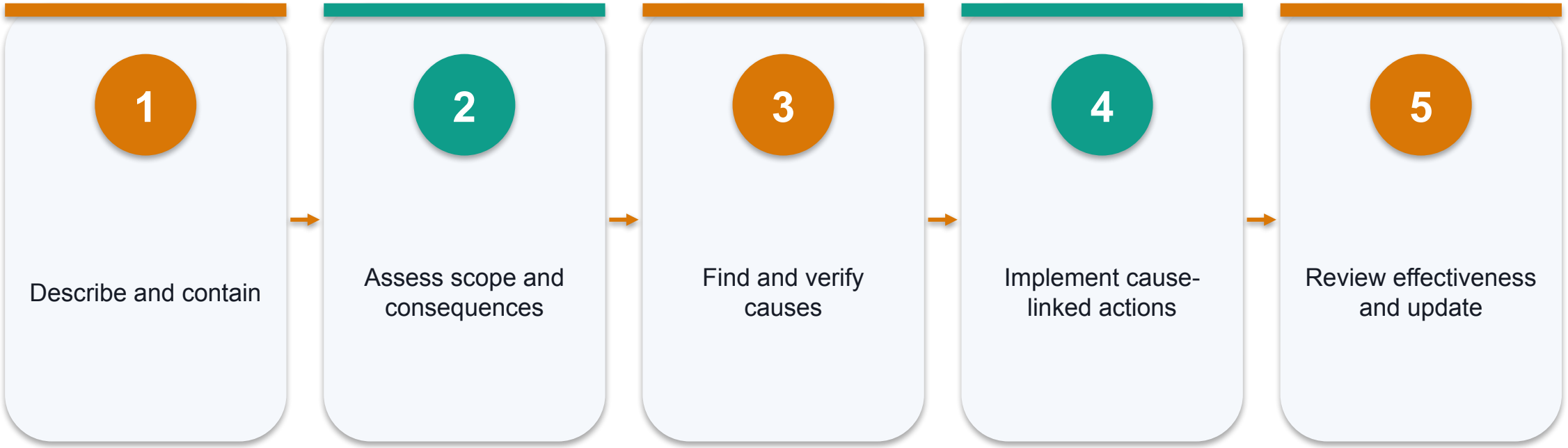
- Control affected scope
- Protect people and receptors
- Limit consequences

Corrective action

- Eliminate verified cause
- Prevent recurrence elsewhere
- System focus

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CAPA logic



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Root cause: evidence before certainty

Useful methods

1. 5 Whys for causal chain
2. Barrier analysis for failed defences
3. Process map for interfaces
4. Data stratification for patterns

Reasoning traps

1. Stop at human error
2. Choose a favourite cause
3. Confuse correlation with cause
4. Skip verification against evidence

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Corrective-action quality test

1. **Cause linked**
   Each action addresses a verified cause.
2. **Owned**
   One accountable owner has authority and resources.
3. **Timed**
   Due date reflects risk and implementation reality.
4. **Systemic**
   Similar aspects, shifts and providers are considered.
5. **Measured**
   Effectiveness criteria are defined before closure.
6. **Sustained**
   Documents, competence and controls are updated.

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VERIFY THE RESULT

Effectiveness is not completion

Implementation evidence

1. Action implemented as planned
2. Evidence shows the new control operates
3. People demonstrate competence
4. Documents and systems are current

Effectiveness evidence

1. Problem does not recur over a meaningful interval
2. Environmental trend moves toward target
3. Audit sample confirms sustained control
4. No impact is transferred elsewhere

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ISO 14001:2026 transition gap review

1 **Context**
Broader environmental conditions and interested-party implications.

2 **Leadership**
Accountability and support for all relevant roles.

3 **Planning**
Clearer risks/opportunities and new Clause 6.3 change management.

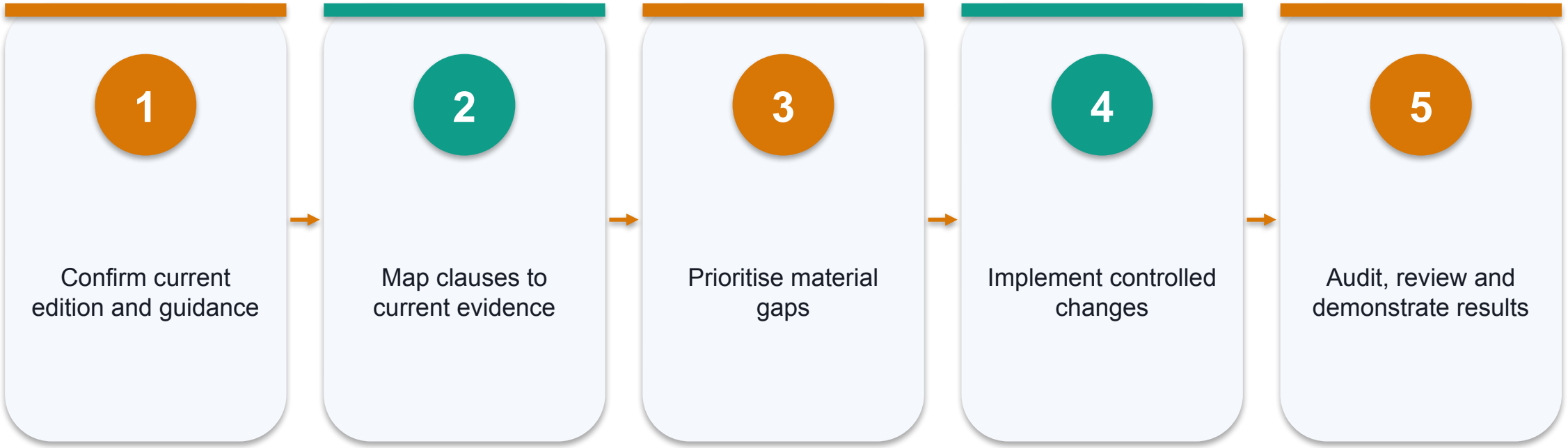
4 **Operation**
Externally provided processes/products/services and life-cycle controls.

5 **Emergency**
All identified potential emergency situations and learning.

6 **Governance**
Audit programme objectives, management-review results and measurable outcomes.

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Transition without losing control



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Close Corrective Action and Plan the 2026 Transition

ACTIVITY 8

Close the spill-response weakness and convert ISO 14001:2026 gaps into a prioritised transition roadmap.

EXPECTED OUTPUT

An NCR/CAPA record, effectiveness review, continual-improvement report and controlled transition plan.

Resources: Corrective Action and Transition Workbook (Excel), report-quality checklist PDF

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ACTIVITY 8 | DECISION PROMPT

Close Corrective Action and Plan the 2026 Transition



The tabletop showed that drain isolation ownership was unclear and contractor contact data was obsolete. Leadership also needs a structured gap review for the 2026 edition, including value-chain oversight and planned change.

Decide what evidence you need, what method fits, and how you will know the result is credible.

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ACTIVITY 8 | VERIFY

Close Corrective Action and Plan the 2026 Transition

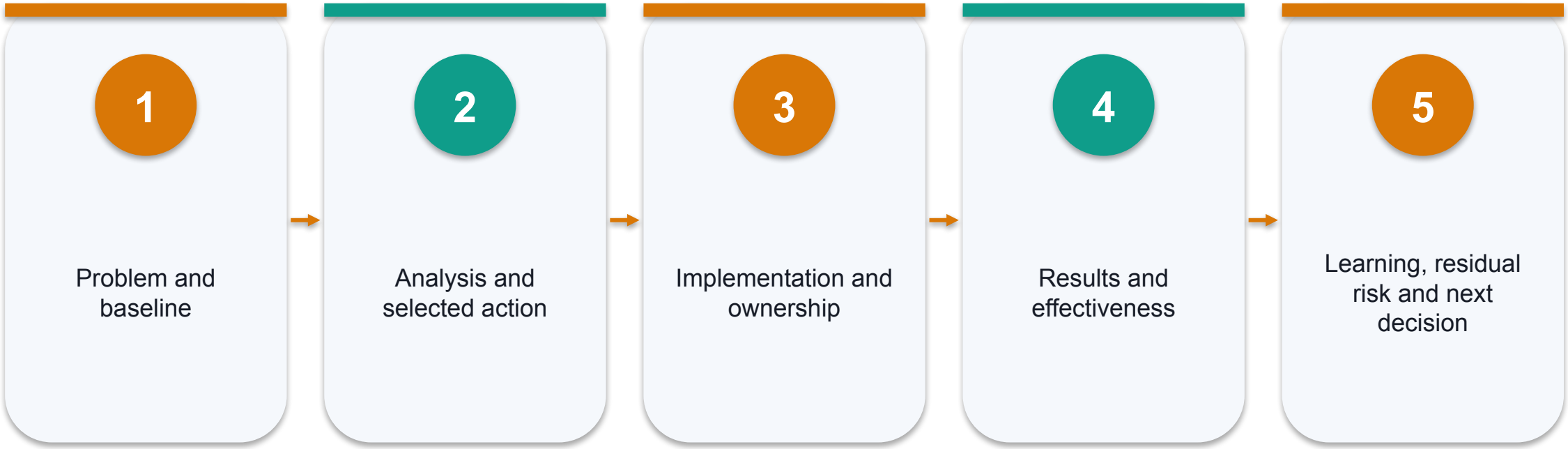
CHECK YOUR EVIDENCE

Actions address verified causes, transition priorities are traceable to current requirements, effectiveness is measured after an appropriate interval, and a reviewer can reconstruct decisions, ownership, evidence and residual risk.

Detailed steps, scenario exhibits and templates are in the Learner Guide and activity folder.

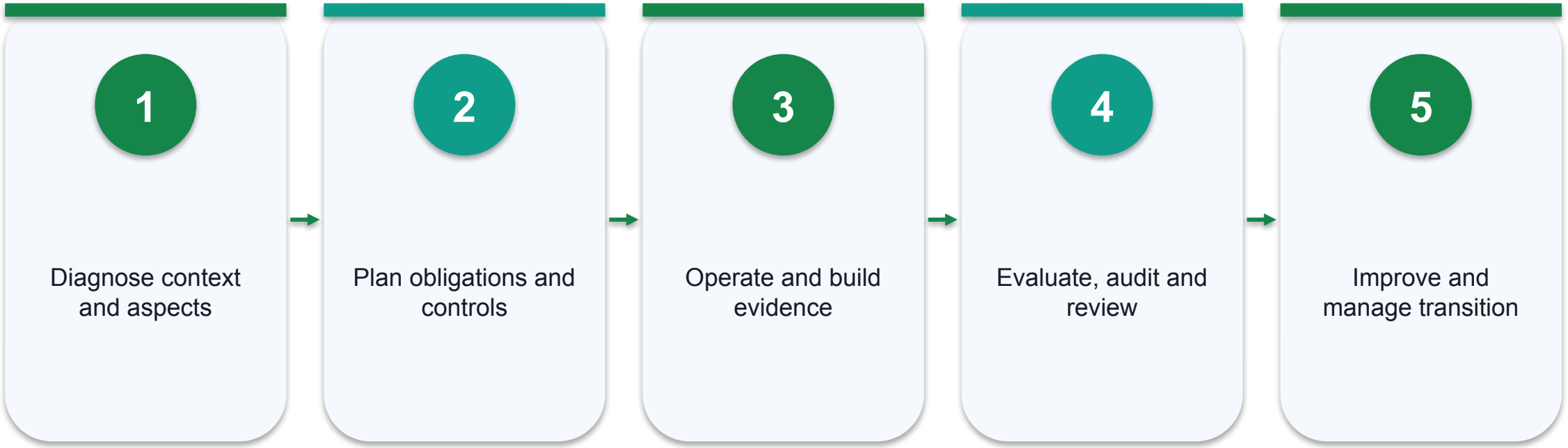
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Improvement report storyline



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From course to workplace



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COURSE SUMMARY

What you can now do

1. **Identify**
   Find environmental hazards, aspects and impacts.
2. **Assess**
   Prioritise significance, risks and opportunities.
3. **Prepare**
   Plan incident action, escalation and reporting.
4. **Control**
   Design life-cycle operational and competence controls.
5. **Evaluate**
   Use performance, compliance and audit evidence.
6. **Improve**
   Close CAPA and manage the 2026 transition.

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Primary and professional references

Course page - duration, outcomes, topics and assessment
Open source →

ISO 14001:2026 standard status
Open source →

ISO 14001:2026 publication announcement
Open source →

ISO 14001:2026 changes explained
Open source →

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REFERENCES

Sources and further reading

BSI ISO 14001:2026 key changes and guidance
Open source →

RRC certification, requirements and compliance guide
Open source →

TaxTMI ISO 14001 EMS overview
Open source →

Complaion ISO 14001 EMS guide
Open source →

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REFERENCES

Sources and further reading

SCMEP ISO 14001 environmental management guide
Open source →

ImpactBuying ISO 14001 guide
Open source →

CTO Framework ISO 14001 requirements map
Open source →

Presencis clause 6.1 guide
Open source →

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REFERENCES

Sources and further reading

Sandland Packaging EMS implementation example
Open source →

CIVAC EMS implementation roadmap
Open source →

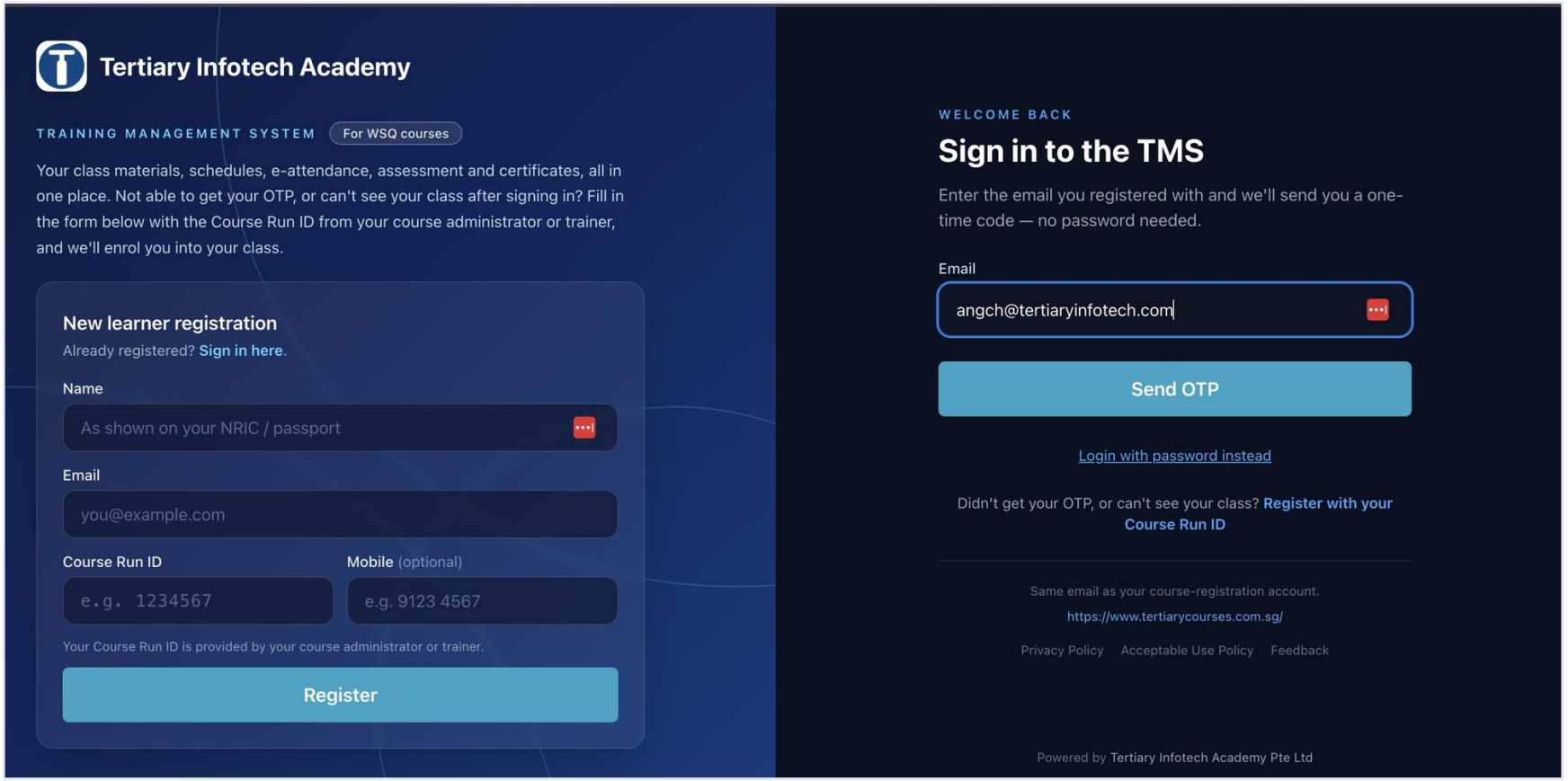
Leadity ISO 14001 requirements guide
Open source →

Singapore NEA environmental legislation portal
Open source →

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COURSE PORTAL

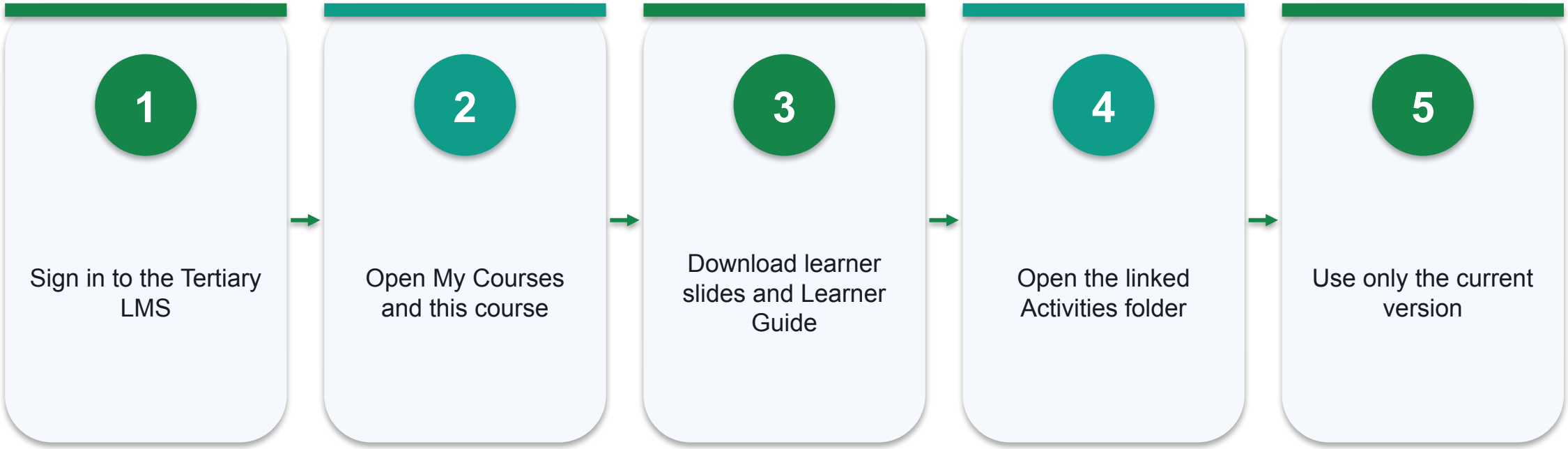
Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Download course material from the LMS



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Assessment reminder

1. Written Assessment (WA/SAQ) - 6 open-ended questions, 1 hour, open book.
2. Case Study (CS) - 4 open-ended scenario questions, 1 hour, open book.
3. Complete Assessment digital attendance before starting.
4. Submit answers on the LMS and sign the Assessment Summary Record.

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Assessment flow



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Digital Attendance (Mandatory)

1. Complete the Assessment digital attendance using the SSG QR code.
2. Confirm your submission before starting the assessment.
3. Attendance evidence is separate from assessment submission.
4. Ask the trainer immediately if confirmation does not appear.

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THANK YOU

Make environmental performance visible.

Use evidence, ownership and continual improvement to turn commitments into controlled, measurable results.

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